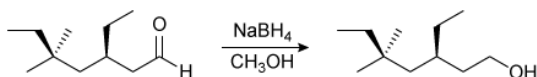


What is the IUPAC name of the product in the reaction below?

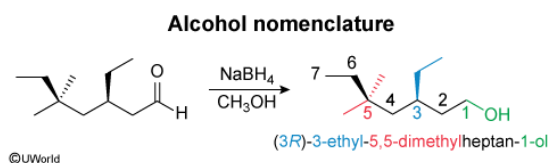


- ☐ A. (3S)-3-ethyl-5,5-dimethylheptan-1-ol [17%]
- ✓ ☒ B. (3R)-3-ethyl-5,5-dimethylheptan-1-ol [67%]
- ☐ C. (3R,5S)-3,5-diethyl-5-methylhexan-1-ol [12%]
- ☐ D. (5R)-5-ethyl-3,3-dimethylheptan-7-ol [2%]

Correct

67% answered correctly

### Explanation:



**Alcohols** are named according to systematic guidelines from the IUPAC. For molecules in which the **highest priority functional group** is the **hydroxyl group** (–OH), the parent chain is identified as the longest continuous carbon chain containing the hydroxyl group. The **prefix** corresponds to the number of carbon atoms in the parent chain, and the **suffix –ol** corresponds to the alcohol functional group. The parent chain is numbered to give the hydroxyl group the lowest number possible, and the location of the hydroxyl group is indicated immediately before the suffix (eg, if the hydroxyl group is on carbon 2 in a three-carbon alcohol, the molecule's name is propan-2-ol).

Substituents are listed before the parent chain using a number to designate its position of attachment to the parent chain. **Stereocenters** (*R/S* or *E/Z*) and their location in the parent chain are given in parentheses at the beginning of the compound name.

The product of the reaction in this question is an alcohol. The parent chain contains seven carbons (prefix *hept–*) and is numbered so the hydroxyl group is given the lowest position number possible (carbon 1, in this case). There is an ethyl substituent on carbon 3 (3-ethyl) and two methyl substituents on carbon 5 (5,5-dimethyl). Carbon 3 is a stereocenter in which the substituents with **priority 1, 2, and 3** are in a clockwise arrangement. Because the lowest-priority group is oriented behind the plane, carbon 3 has an **R configuration**.

Therefore, the IUPAC name of the alcohol in the given reaction is **(3R)-3-ethyl-5,5-dimethylheptan-1-ol**.

**(Choice A)** (3S)-3-ethyl-5,5-dimethylheptan-1-ol incorrectly classifies the stereocenter at carbon 3 as *S* instead of *R*.

**(Choice C)** (3R,5S)-3,5-diethyl-5-methylhexan-1-ol **incorrectly identifies** the parent chain and carbon 5 of the parent chain as a stereocenter.

**(Choice D)** (5*R*)-5-ethyl-3,3-dimethylheptan-7-ol **incorrectly numbers** the parent chain so that the hydroxyl group has the *highest* instead of the lowest possible position.

**Educational objective:**

Alcohols are molecules that contain a hydroxyl group (–OH) and are named using the suffix –*ol*. The prefix in the name indicates the length of the parent chain (ie, the longest continuous carbon chain containing the hydroxyl group). Substituents and stereocenters are listed preceding the parent chain name.

**Time spent:**0 sec

**Subject:**  
Organic Chemistry

**QID:**403889

**System:**  
Functional Groups and Their Reactions